



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2028	B.TECH	IIT Kharagpur	8.11 / 10
2023	Class XII, CBSE	Young Phoenix Public School, Bhubaneswar	92%
2021	Class X, ICSE	Cambridge School, Cuttack	92%

INTERNSHIPS

Software Developer Intern ResearchFundamental Pvt. Ltd.	Jan 2026 - Present
- Extracted 11.7M daily OHLCV bars across 4,391 BSE tickers, cross-verifying 3 sources to flag 218 stocks with unadjusted corporate actions - Built the OEM/FADA analytics module computing growth, margins and volumes benchmarked across 50+ India-market OEMs from FADA - Automated GitHub issue creation via form submissions and built an agent pipeline where Claude raised fix PRs gated behind team's review - Developed interactive financial model and chart views in React with MUI X Charts for 3000+ companies, with projections and trends	

COMPETITION/CONFERENCE

InterIIT Tech Meet 14.0 Pathway - Silver Medal Contingent Gold	Nov 2025 - Dec 2025
- Constructed a polymorphic 37-pattern candlestick detection engine evaluating 1-hour OHLCV data formations across 80 US stock tickers - Scheduled nightly cron-driven batch workflows running LLM reasoning over Postgres-backed historical candlestick data for explanations - Contributed to a 15-service containerized platform using host-mode Docker Compose spanning FastAPI, PostgreSQL, Redis, and Kafka - Broadcast real-time candlestick pattern detection and LLM reasoning over WebSockets using Kafka event streams to the frontend chart	

PROJECTS

Vylos: Containerized PaaS Deployment Platform	May 2026 - Jun 2026
- Orchestrated full container lifecycle via Docker SDK, deploying GitHub repos to live subdomains, scaling to 8 containers on one EC2 node - Configured a dynamic Nginx reverse-proxy generator for per-app subdomain routing, auto-reloading configs on EC2 via Docker Compose - Secured all FastAPI routes with stateless JWT session tokens issued behind OAuth 2.0 login via both GitHub and Google identity providers - Streamed real-time container logs to the frontend dashboard via persistent SSE connection, providing users live build and runtime visibility	
Wind Optimal Flight Route Planner	Feb 2026 - Apr 2026
- Built a 3D (lat/lon/altitude) graph optimizer minimizing fuel burn through real NOAA GFS winds, saving 2.6 - 6.9% vs the great-circle route - Implemented an A* algorithm using a provably admissible tailwind heuristic, expanding 42% fewer nodes than Dijkstra at equal cost - Modeled a turn-radius-constrained hybrid A* algorithm over a 4-D state space using aircraft banked-turn physics to ensure flyable routes - Compressed wind data transfer by ~13,700x (490 MB to 37 KB) via server-side GRIB-filter subsetting, caching grids inside Redis on FastAPI	
Query Management System	Dec 2025 - Jan 2026
- Segmented a role-based access platform across 5 distinct teams (management, sponsorship, tech, PR, design) with RBAC permissions - Allocated incoming tickets via LLM-driven auto-assignment, matching free-text queries against a JSON knowledge base for fast resolution - Instituted an admin-controlled workflow gating escalation, reassignment, and status updates with automated email upon assignment - Monitored assigned and unassigned ticket metrics alongside per-head workload activity on the interactive analytics dashboard for admins	

SKILLS AND EXPERTISE

Languages & Tools: JavaScript TypeScript Python C/C++ HTML CSS Git GitHub Linux VSCode Postman Jupyter Notebook
Frameworks & Libraries: Express.js Node.js Next.js React FastAPI MongoDB Numpy Pandas Matplotlib Axios C++ STL
DevOps, Cloud & Databases: Docker Docker Compose Nginx Kafka AWS (EC2, S3) MongoDB PostgreSQL Redis

COURSEWORK INFORMATION

Computer Science: Programming and Data Structures Essentials of Machine Learning Fundamentals of Embedded Control and Software
Mathematics: Linear Algebra, Numerical & Complex Analysis Advanced Calculus Integral Transforms Advanced Computational Methods

POSITIONS OF RESPONSIBILITY

Tech Head National Students' Space Challenge IIT Kharagpur	May 2025 - Present
- Built the website backend in Node.js and Express.js, powering 4500+ registrations across 300+ teams for 10+ events and 5+ workshops - Owned 5 society websites end-to-end across 5 GitHub organizations and 10+ repositories, from MERN development to Vercel CLI deploys - Ran on-ground operations for 500+ attendees at NSSC 2025, covering registration, accommodation, merchandise, and coupon distribution	
Executive Member Space Technology Students' Society Kalpana Chawla Space Technology Cell IIT Kharagpur	Feb 2025 - Present
- Recruited, trained, and now lead the tech team building the NSSC 2026 website portal, owning the stack, task allocation, and code review - Cultivated frontend development proficiency across 13 junior members with hands-on build tasks for the NSSC 2026 production codebase - Ran an internal hackathon for the team, setting build challenges that sharpened hands-on development and the team's problem-solving	

EXTRA CURRICULAR ACTIVITIES

- Mentored Class 9-10 students at Allen Career Institute on academic paths, exam strategy, subject planning, and long term career decisions - Volunteered with NSS, mobilizing student and faculty donations to fund welfare, education, and disaster-relief drives for communities - Secured Gold in Illumination and Silver in Rangoli at Diwali inter-hall contest, representing Madan Mohan Malviya Hall of Residence	
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